

Julius Zhang

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I am interested in identifying structures and developing new primitives for practical problems.

EXPERIENCE

Columbia University - Conduct research and attend seminars in the Department of Mathematics. - Homotopy theory, arithmetic geometry, cryptography, security, and other topics.	Researcher, 2026.6 - Present
a16z - Building Jolt, open-source zero-knowledge virtual machine in Rust. - Implemented and benchmarked torus compression; proved optimality with a conceptual reframing via Hilbert 90. - Audited proof systems security: verify correctness of elliptic-curve optimizations and formalize folklore results into rigorous proofs. - Evaluated lattice-based cryptography parameters against known attacks under various security regimes and performance constraints. - Performance engineering across the proving stack, including sum-check optimization.	Researcher, 2025.4 - Present
Millennium Convex optimization and robust statistical estimation for real-time trading systems.	Researcher, 2024.6 - 2025.4
Stellar Development Foundation - Byzantine fault tolerance, C++, stellar-core. - Improved maximum transaction rate per second by 17% via changes to the consensus protocol, reducing number of communication rounds in overlay network. Formal verification in TLA+.	Consultant, 2023.6 - 2023.9
Distributed Systems and TCS Research, Stanford University - Under David Mazières, research in general open membership Byzantine fault tolerance protocols (Github), with applications to certificate authority management and automatic password recovery with hardware security modules. - Research PCP under Aviad Rubinfeld, reconstructed proof of the 2-to-2 Games Conjecture. Proved a result about Convex Continuous Class Complexity.	Researcher, 2023.9 - 2024.6
Topology Research, Stanford University - Floer homotopy theory under Mohammed Abouzaid; related the complex cobordism Gysin map to the Grothendieck residue symbol and derived a recursive formula for Gromov–Witten quantum product corrections. - Research under Ciprian Manolescu. Floer theory, Seiberg-Witten, knot invariants. Produced new potential counterexamples for the smooth Poincare conjecture.	Researcher, 2020.7 - 2022.5

SELECTED PAPERS AND MANUSCRIPTS

- Julius Zhang. *The Trace-Zero Subgroup and the Relative Trace for BN254*, 2026. [\[PDF\]](#)
- Julius Zhang. *Complex Cobordism and Formal Group Law*, Manuscript (Undergraduate Honours Thesis), Stanford University, 2024. [\[PDF\]](#)

TALKS

- p*-Ordinary Cohomology for $SL(2)$ over Number Fields, Hida Theory Seminar, Columbia University, 2026.
- Construction of Seiberg-Witten-Floer stable homotopy types, Homotopy Theory and Manifolds Seminar, Columbia University, 2026.

EDUCATION

Courant Institute of Mathematical Sciences, New York University , computer science <i>PhD</i> . Verifiable systems and theoretical computer science.	starting 2026.9
Stanford University , computer science <i>M.S.</i> and mathematics <i>B.A.</i> General Wang Yaowu Scholarship.	2019.9 - 2024.6

SKILLS

Computer Science: Rust, C/C++, Python, formal verification, convex optimization, distributed systems, cryptographic protocol design, performance engineering.

Mathematics: number theory, algebraic geometry, symplectic topology, homotopy theory, Floer theory, gauge theory, concentration inequalities.